微积分 A2（留学生） **姓名:** ____________________

2025 S

期中考试

2025.4.13

考试时间: 3 小时 **学生 ID:** ____________________

This exam consists of 16 pages (including this page) and 7 major questions. Please check if any pages are missing, and write your student ID and name on this page as well as on the first page of your answer sheet. It is also recommended that you write the initials of your name on the top right corner of every page of the answer sheet, to prevent pages from getting lost.

This exam is open book. You may consult any printed materials, but no electronic devices—including electronic watches and calculators—are allowed. Any form of cheating is strictly prohibited and will be subject to disciplinary action.

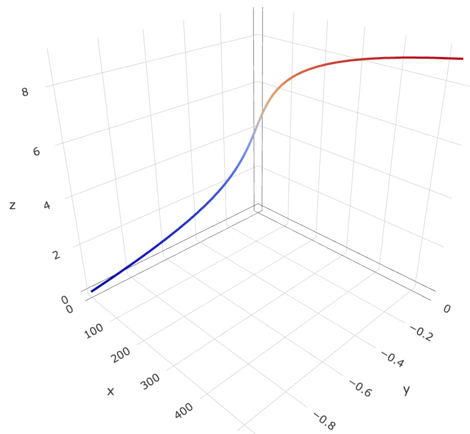
Please write your answers on the designated **answer sheet** and follow these rules:

- **Please explain your answers.** Do not simply write down an answer; every response must include at least one sentence of explanation.
- **Please write clearly and neatly.** If the examiner cannot understand your writing, you may receive no points.
- **We award points for the steps.** As long as you demonstrate an understanding of the process, you will receive some points. Minor calculation errors will not be heavily penalized.
- Please note that the blank spaces on the exam paper may be used as scratch paper, but they will not be graded. Write your answers on the designated **answer sheet**.

大题	Points	Score
1	16	
2	13	
3	12	
4	9	
5	17	
6	5	
7	2	
Total:	74	

1. I want to design a super awesome water slide. Note that this is the intersection curve of the two surfaces $xy + 1 = 0$ and $x = \mathrm{e}^{\frac{z}{\sqrt{2}}}$. One possible parametrization is

$$\gamma(t) = \begin{bmatrix} \mathrm{e}^t \\ -\mathrm{e}^{-t} \\ \sqrt{2}t \end{bmatrix}.$$



(a) (5 points) At the point $(1, -1, 0)$, find the normal vectors of the surfaces $xy + 1 = 0$ and $x = \mathrm{e}^{\frac{z}{\sqrt{2}}}$, and compute their cross product. What is the relation between this cross product and the curve?

(b) (4 points) Suppose we want to measure the length of the water slide. Compute the length of the curve for $t \in [-1, 1]$.

(c) (4 points) For the function $f(x, y, z) = xy + z^2$, compute $\frac{\mathrm{d}f}{\mathrm{d}t}$.

(d) (3 points) (Hard) Compute the vectors $\boldsymbol{T}, \boldsymbol{N}, \boldsymbol{B}$ of the curve at the point $(1, -1, 0)$. (Hint: One method is to find $\boldsymbol{N}$ via the Frenet decomposition of the acceleration $\gamma''(t)$.)

Answer:

Note that:

$$\gamma'(t) = \begin{bmatrix} e^t \\ e^{-t} \\ \sqrt{2} \end{bmatrix} \implies \gamma'(0) = \begin{bmatrix} 1 \\ 1 \\ \sqrt{2} \end{bmatrix}.$$

$$\gamma''(t) = \begin{bmatrix} e^t \\ -e^{-t} \\ 0 \end{bmatrix} \implies \gamma''(0) = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}.$$

$$\gamma'''(t) = \begin{bmatrix} e^t \\ e^{-t} \\ 0 \end{bmatrix} \implies \gamma'''(0) = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

From our calculations, we find:

$$||\gamma'(t)|| = e^t + e^{-t} \implies ||\gamma'(0)|| = 2.$$

$$||\gamma'(0) \times \gamma''(0)|| = \left|\left| \begin{bmatrix} \sqrt{2} \\ \sqrt{2} \\ -2 \end{bmatrix} \right|\right| = 2\sqrt{2}.$$

$$\det(\gamma'(0), \gamma''(0), \gamma'''(0)) = 2\sqrt{2}.$$

1. The normal vector of $xy+1=0$ is $\nabla(xy+1) = \begin{bmatrix} y \\ x \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$, and the normal vector of $x = e^{\frac{z}{\sqrt{2}}}$ is $\nabla(x - e^{\frac{z}{\sqrt{2}}}) = \begin{bmatrix} 1 \\ 0 \\ -\frac{1}{\sqrt{2}} e^{\frac{z}{\sqrt{2}}} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$. The result of the cross product is $\begin{bmatrix} -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ -1 \end{bmatrix}$, which is parallel to the tangent vector of the curve.
2. $$\int_{-1}^{1} ||\gamma'(t)||\, dt = \int_{-1}^{1} (e^t + e^{-t})\, dt = (e^t - e^{-t})\Big|_{-1}^{1} = 2e - 2e^{-1}.$$
3. Note that $f = -1 + 2t^2$, so $\frac{df}{dt} = 4t$.
4. We directly calculate $\mathbf{T} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$. Since $\gamma''(0) = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ is already orthogonal to $\mathbf{T}$, and the Frenet decomposition of acceleration implies that $\gamma''(0)$ is a linear combination of $\mathbf{T}$ and $\mathbf{N}$, it follows that $\gamma''(0)$ is a positive multiple of

$\mathbf{N}$. Therefore, $\mathbf{N} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}$. Finally, the cross product gives us $\mathbf{B} = \mathbf{T} \times \mathbf{N} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$.

2. Suppose we want to solve for the intersection points on the plane between the unit circle and the line $x = y$. (Assume we do not know how to compute it directly.) Define

$$\boldsymbol{F}(x, y) = \begin{bmatrix} x^2 + y^2 - 1 \\ x - y \end{bmatrix}.$$

Then we want to solve $\boldsymbol{F} = \mathbf{0}$. We first arbitrarily guess the answer is $(1, 1)$, and then hope to use Newton's method to solve it.

(a) (4 points) Compute the Jacobian matrix at $(1, 1)$, and explain why the function is differentiable at this point (i.e. prove that the total derivative matrix exists here).

(b) (2 points) Estimate $\boldsymbol{F}(1.001, 1.002)$.

(c) (4 points) Compute the partial derivatives of $\boldsymbol{F}$ with respect to the polar coordinates r, θ, namely $D_r\boldsymbol{F}, D_\theta\boldsymbol{F}$.

(d) (3 points) Starting from the point $(1, 1)$, take one step using Newton's method. What is the next point? (Is it closer to the answer?)

Answer:

1. The Jacobian is $[D\mathbf{F}] = \begin{bmatrix} 2x & 2y \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 1 & -1 \end{bmatrix}$. It is differentiable because all partial derivatives are continuous.
2. $$\mathbf{F}(1.001, 1.002) \approx \mathbf{F}(1,1) + \begin{bmatrix} 2 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0.001 \\ 0.002 \end{bmatrix} \approx \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.006 \\ -0.001 \end{bmatrix} = \begin{bmatrix} 1.006 \\ -0.001 \end{bmatrix}.$$
3. Expressing $\mathbf{F}$ in polar coordinates: $\mathbf{F} = \begin{bmatrix} r^2 - 1 \\ r(\cos\theta - \sin\theta) \end{bmatrix}$. Thus, $D_r\mathbf{F} = \begin{bmatrix} 2r \\ \cos\theta - \sin\theta \end{bmatrix}$ and $D_\theta\mathbf{F} = \begin{bmatrix} 0 \\ -r(\sin\theta + \cos\theta) \end{bmatrix}$.
4. The next step is:
$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 & 2 \\ 1 & -1 \end{bmatrix}^{-1} \mathbf{F}(1,1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} 1 & 2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1/4 \\ 1/4 \end{bmatrix} = \begin{bmatrix} 3/4 \\ 3/4 \end{bmatrix}.$$

3. We demonstrate an application of the Implicit Function Theorem in chemical engineering. In a chemical reaction, assume the concentrations of three reactants are x, y, z, and the temperature is T. Then assume that the relationship among these four quantities necessarily satisfies the equation

$$x + y^2 + z^3 - \mathrm{e}^T = 0.$$

(a) (3 points) At $(x, y, z, T) = (1, 1, 1, \ln 3)$, compute the normal vector of this three-dimensional surface.

(b) (2 points) At $(x, y, z, T) = (1, 1, 1, \ln 3)$, treat T as a function of x, y, z, and compute the partial derivative $\frac{\partial T}{\partial y}$. (How to adjust the concentration y to control the temperature T)

(c) (2 points) At $(x, y, z, T) = (1, 1, 1, \ln 3)$, treat y as a function of x, z, T, and compute the partial derivative $\frac{\partial y}{\partial T}$. (How to adjust the temperature to control the concentration y)

(d) (2 points) Please prove or provide a counterexample: when $y \neq 0$, is it always true that $\frac{\partial T}{\partial y} \frac{\partial y}{\partial T} = 1$?

(e) (3 points) At $(x, y, z, T) = (1, 1, 1, \ln 3)$, provide a tangent vector in the direction of steepest ascent of the graph of the function $T(x, y, z)$ (note: the tangent vector is in $\mathbb{R}^4$!).

Answer:

1. The gradient $\nabla(x+y^2+z^3-e^T) = \begin{bmatrix} 1 \\ 2y \\ 3z^2 \\ -e^T \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ -3 \end{bmatrix}$ is the normal vector.
2. This is $\frac{2}{3}$.
3. This is $\frac{3}{2}$.
4. It always holds, because by the Implicit Function Theorem, when $f = x + y^2 + z^3 - e^T$, the former is $-\frac{D_y f}{D_T f}$ and the latter is $-\frac{D_T f}{D_y f}$. Note that when $y \neq 0$, we always have $D_T f, D_y f \neq 0$.
5. Note that $\nabla T = \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \\ 1 \end{bmatrix}$, therefore a tangent vector in the direction of steepest ascent is $\begin{bmatrix} 1 \\ 2 \\ 3 \\ \frac{14}{3} \end{bmatrix}$.

4. Consider a vector field describing water flow

$$\boldsymbol{F}(x,y,z) = \begin{bmatrix} x - y - xz \\ x + y - yz \\ z - x^2 - y^2 \end{bmatrix}.$$

(a) (5 points) Compute its divergence and curl at the point (x,y,z).

(b) (2 points) Note that $\boldsymbol{F}(\mathbf{0}) = \mathbf{0}$. Compute the total derivative matrix $[D\boldsymbol{F}(\mathbf{0})]$ at this point, and determine: is this a source, a sink, or neither?

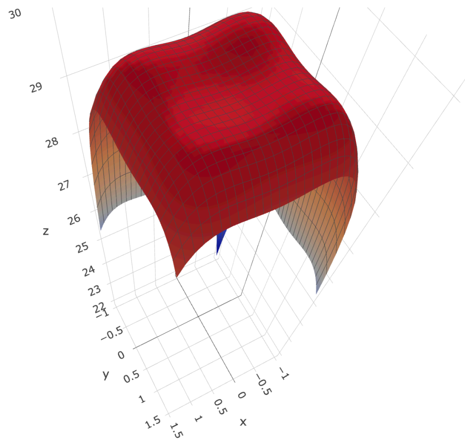
(c) (2 points) The Inverse Function Theorem guarantees that $\boldsymbol{F}$ is invertible on a neighborhood of $\mathbf{0}$. Compute the total derivative matrix $[D\boldsymbol{F}^{-1}(\mathbf{0})]$ of the inverse mapping $\boldsymbol{F}^{-1}$ at $\mathbf{0}$ in that neighborhood.

Answer:

1. $\nabla \cdot \mathbf{F} = (1-z)+(1-z)+1 = 3-2z$. The curl is $\nabla \times \mathbf{F} = \begin{bmatrix} -2y+y \\ -x+2x \\ 1+1 \end{bmatrix} = \begin{bmatrix} -y \\ x \\ 2 \end{bmatrix}$.
2. $[D\mathbf{F}(\mathbf{0})] = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Note that the eigenvalues are $1, 1+i, 1-i$. Since the real parts of all eigenvalues are greater than zero, this is a source point.
3. The Inverse Function Theorem guarantees invertibility in a neighborhood of the origin. The Jacobian of the inverse mapping is: $[D\mathbf{F}^{-1}(\mathbf{0})] = [D\mathbf{F}(\mathbf{0})]^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

5. In a city, the downtown area is often busier than the suburbs, and therefore has a higher temperature. This is called the urban heat island effect. Suppose the temperature distribution of a city is

$$T(x, y) = 30 - (x^4 + y^4) + 2(x^3 + y^3) - (x^2 + y^2).$$



(a) (4 points) Find all points where the gradient is zero.

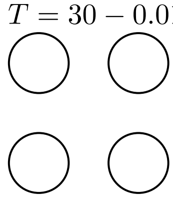
(b) (6 points) Compute the Hessian matrix at these points, and determine which are local maxima, which are local minima, and which are saddle points.

(c) (4 points) In the city, the second ring road is the square formed by the four edges $x = 1/2$, $y = 1/2$, $x = 3/2$, $y = 3/2$. Find the local maximum points on the second ring road. (To simplify the computation, you only need to examine the two line segments corresponding to $x = 1/2$ and $y = 1/2$ in the square, but do not forget the endpoints.)

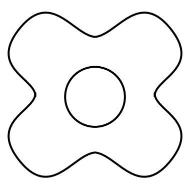
(d) (3 points) Please sketch the three level sets $T = 30 - 0.01$, $T = 30 - 0.1$, $T = 30 - 0.2$.

Answer:

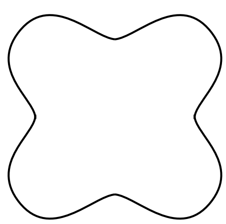
1. The critical points occur where $x, y \in \{0, \frac{1}{2}, 1\}$. There are 9 points in total from all permutations.
2. Classification using the Hessian:
   - **Local Maxima:** The four permutations of $x, y \in \{0, 1\}$.
   - **Local Minimum:** The point $(1/2, 1/2)$.
   - **Saddle Points:** The remaining four points (where one coordinate is $1/2$ and the other is 0 or 1).
3. The maxima on the segments occur at $(1/2, 1)$ and $(1, 1/2)$.
4. - $T = 30 - 0.01$ **level set:** Four separate circles around the peaks.

   - $T = 30 - 0.1$ **level set:** An outer four-petaled flower shape with an inner circle.

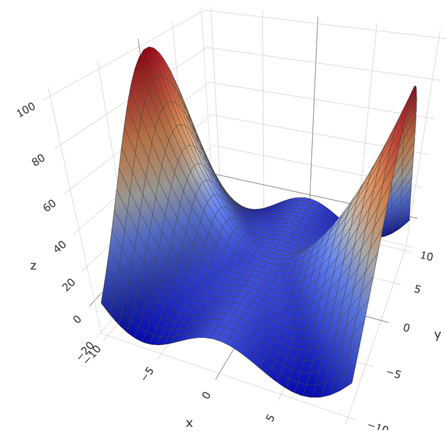
     

   - $T = 30 - 0.2$ **level set:** A single four-petaled flower boundary.

6. When a rubber material is under stress, it becomes deformed. Define the function

$$f(x,y)=\begin{cases}\dfrac{x^2\left(x^2-y^2\right)}{x^2+y^2}, & (x,y)\neq(0,0),\\ 0, & (x,y)=(0,0),\end{cases}$$



We have computed its partial derivatives (and mixed partial derivatives) away from the origin, and the answers are as follows.

$$D_xf(x,y)=\begin{cases}\dfrac{2x\left(x^4+2x^2y^2-y^4\right)}{(x^2+y^2)^2}, & (x,y)\neq(0,0),\\ ???, & (x,y)=(0,0),\end{cases}$$

$$D_yf(x,y)=\begin{cases}-\dfrac{4x^4y}{(x^2+y^2)^2}, & (x,y)\neq(0,0),\\ ???, & (x,y)=(0,0),\end{cases}$$

$$D_xD_yf(x,y)=\begin{cases}-\dfrac{16x^3y^3}{(x^2+y^2)^3}, & (x,y)\neq(0,0),\\ ???, & (x,y)=(0,0),\end{cases}$$

$$D_yD_xf(x,y)=\begin{cases}-\dfrac{16x^3y^3}{(x^2+y^2)^3}, & (x,y)\neq(0,0),\\ ???, & (x,y)=(0,0),\end{cases}$$

(a) (3 points) Is f continuous? Is D_xf continuous? Is D_xD_yf continuous?

(b) (1 point) Do we have $D_xD_yf(0,0)=D_yD_xf(0,0)$?

(c) (1 point) Let $\boldsymbol{v}=(1,1)$. Then, do we have $D_{\boldsymbol{v}}D_xf(0,0)=D_xD_{\boldsymbol{v}}f(0,0)$?

Answer:

1. Both the function and its first partial derivatives are continuous (note that the inequality $|\frac{x^2-y^2}{x^2+y^2}| \leq 1$ allows for a squeeze theorem argument at the origin). However, the second-order mixed partial derivatives are discontinuous (this can be shown by testing the path $y = kx$ as it approaches the origin, where the limit will depend on k).
2. Yes, both are equal to zero.
3. The former ($D_{\mathbf{v}}D_x f(0,0)$) is 1, while the latter ($D_x D_{\mathbf{v}} f(0,0)$) is 2.

7. (Challenge Problem) Compute the derivative problems for the following abstract mappings.
   (a) (1 point) $\boldsymbol{f}(\boldsymbol{x}, \boldsymbol{y}) = \boldsymbol{x}\boldsymbol{y}^{\mathrm{T}}$, which maps elements from $\mathbb{R}^{2n}$ to an $n \times n$ matrix. Compute the directional derivative $D_{(\boldsymbol{x},\boldsymbol{y})}\boldsymbol{f}(\boldsymbol{a}, \boldsymbol{b})$.

   (b) (1 point) Let V be the space of all infinitely differentiable real-valued functions of one variable, and let $\frac{\mathrm{d}}{\mathrm{d}x} : V \to V$ be the differentiation operator that maps each function to its derivative. Compute the directional derivative $D_g \frac{\mathrm{d}}{\mathrm{d}x}(f)$.

Answer:

1. $D_{(\mathbf{x},\mathbf{y})}\mathbf{f}(\mathbf{a},\mathbf{b}) = \mathbf{x}\mathbf{b}^\top + \mathbf{a}\mathbf{y}^\top$.
2. $D_g \frac{d}{dx}(f) = \frac{dg}{dx}$. The operator is linear. The derivative of a linear map is the linear map itself. (Because the linear appoximation of a linear map is just the linear map itself.)